

Mark schemes

Q1.

(a) $y = x + 3 + \frac{8}{x^4} = x + 3 + 8x^{-4}$

For $\frac{8}{x^4} = 8x^{-4}$ Pl by correct differentiation

of 3rd term

B1

$$\frac{dy}{dx} = 1 - 32x^{-5} \text{ or } 1 - \frac{32}{x^5}$$

kx^{-5} OE

M1

For either

A1

3

(b) When $x = 1, y = 12$

B1

When $x = 1, \frac{dy}{dx} = 1 - 32 = -31$

Attempt to find value of $\frac{dy}{dx}$ when $x = 1$

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Tangent: $y - 12 = -31(x - 1)$

Only ft on c's answer to (a). Any correct
(ft on c's (a)) form.

A1F

3

(c) $1 - 32x^{-5} = 0$

$$1 - 32x^{-5} = 0 \text{ or c's } \frac{dy}{dx} = 0$$

M1

$$\Rightarrow x^5 = 32$$

Attempt to form $x^n = \text{const} (\neq 0)$. PI by next line

m1

$$\Rightarrow x = 2$$

CSO

A1

(Coordinates of M) (2, 5.5)

CSO

A1

4

(d) (i)

$$\int \left(x + 3 + \frac{8}{x^4} \right) dx$$

$$= \frac{x^2}{2} + 3x - \frac{8}{3}x^{-3} + c$$

Power -3 correctly obtained

M1

$$- \frac{8}{3}x^{-3}$$

A1

$$\frac{x^2}{2} + 3x + c$$

B1

3

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(ii)

$$= \left(2 + 6 - \frac{1}{3} \right) - \left(\frac{1}{2} + 3 - \frac{8}{3} \right)$$

Attempting to calculate $F(2) - F(1)$ where $F(x)$ is c's answer to part (d)(i) provided F is not just the c's integrand $(x + 3 + 8/x^4)$

M1

$$\frac{9}{2} + \frac{7}{3} = \frac{41}{6}$$

OE Accept 6.83 or better provided d(i) used

A1

2

(e) $k = -5.5$

Ft on $-y_M$ from part (c).

B1F

1

[16]

Q2.

(a) $\frac{dy}{dx} = k(x^3 - 1)^5$

Where k is an integer or function of x

M1

$= 6 \times 3x^2(x^3 - 1)^5$ (ISW)

But note

$\frac{dy}{dx} = k(x^3 - 1)^5 + px^2$

M0

Or

$(u = x^3 - 1) \quad (y = u^6)$

$\frac{dy}{du} = 6u^5 \text{ and } \frac{du}{dx} = 3x^2$

M1

$= 6(x^3 - 1)^5 \times 3x^2$

A1

Note

$\frac{dy}{dx} = 6 \times 3x^2(x^3 - 1)^5 + c$

scores M1 A0

(penalise + c in differential once only in paper)

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A1

2

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(b) (i) $\frac{dy}{dx} = \pm x \times \frac{1}{x} \pm \ln x$

Product rule attempted and differential of $\ln x$

M1

$= 1 + \ln x$ (ISW)

A1

2

(ii) $(x = e) \quad y = e$ PI

Must have replaced $\ln e$ by 1

Condone $y = 2.72$ (AWRT)

B1

$$\frac{dy}{dx} = 1 + \ln e (= 2)$$

$\frac{dy}{dx}$
 Correct substitution into their
 But must have scored M1 in (b)(i)

M1

$$y - e = 2(x - e) \text{ or } y = 2x - e \quad \text{OE, ISW}$$

Must have replaced $\ln e$ by 1

A1

3

[7]

Q3.

(a) (i) $\frac{dx}{dt} = 3e^t \quad \frac{dy}{dt} = 2e^{2t} + 2e^{-2t}$

Both derivatives attempted and one correct

M1

Both correct

A1

$t = 0 \quad \text{gradient} = \frac{4}{3}$

$\frac{dy}{dx} = \frac{4}{3}$
 cso Condone

A1

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3

(ii) $y \frac{4}{3} (x - 3) \quad \text{oe}$

ft on non-zero gradient

B1ft

1

(b) $e^{2t} = \frac{x^2}{9} \text{ or } 9e^{2t} = x^2 \text{ or } e^t = \frac{x}{3} \text{ or } e^{2t} = \left(\frac{x}{3}\right)^2$

or $t = \ln\left(\frac{x}{3}\right) \text{ or } 2t = \ln\left(\frac{x^2}{9}\right)$

M1

$$y = \frac{x^2}{9} - \frac{9}{x^2}$$

Equation required

A1

2

[6]

Q4.

(a) $\left(\frac{dV}{dt} = \right) \frac{3t^2}{4} - 3$

one of these terms correct

M1

all correct (no + c etc)

A1

2

(b) (i) $t = 1 \Rightarrow \frac{dV}{dt} = \frac{3}{4} - 3$

substituting $t = 1$ into their $\frac{dV}{dt}$

M1

$$= -2\frac{1}{4}$$

(- 2.25 OE) BUT must have $\frac{dV}{dt}$ correct

A1cso

2

(ii) Volume is decreasing when $t = 1$

must have used $\frac{dV}{dt}$ in (b)(i) or starts again

because $\frac{dV}{dt} < 0$

must state that $\frac{dV}{dt} < 0$ (or $-2\frac{1}{4} < 0$ etc)

ft increasing plus explanation if their $\frac{dV}{dt} > 0$

E1 ✓

1

(c) (i) $\left(\frac{dV}{dt} = 0 \Rightarrow\right) \frac{3t^2}{4} - 3 = 0$

PI by "correct" equation being solved

M1

$$\Rightarrow t^2 = 4$$

obtaining $t = k$ correctly from their $\frac{dV}{dt}$

A1 ✓

$$t = 2$$

withhold if answer left as $t = \pm 2$

A1cso

3

(ii) $\left(\frac{d^2V}{dt^2} =\right) \frac{3t}{2}$

(condone unsimplified) ft their $\frac{dV}{dt}$

B1 ✓

When $t = 2$, $\frac{d^2V}{dt^2} = 3$ or $\frac{d^2V}{dt^2} > 0$

ft their $\frac{d^2V}{dt^2}$ and value of t from (c)(i)

M1

$\Rightarrow$ minimum

A1cso

3

[11]

Q5.

(a) *For either 6 or $6x^0$*

B1

$$\frac{dy}{dx} = 6 - 3x^{\frac{1}{2}}$$

$$Ax^{\frac{3}{2}-1}, A \neq 0 \text{ OE}$$

M1

$$6 - 3x^{\frac{1}{2}} \text{ or } 6 - 3\sqrt{x} \text{ with no '+c'}$$

[If unsimplified here, A1 can be awarded retrospectively if correct simplified expression is seen explicitly in (b)(i).]

A1

3

(b) (i) $6 - 3x^{\frac{1}{2}} = 0$

Equating c's $\frac{dy}{dx}$ to 0 PI by correct ft
rearrangement of c's $dy/dx = 0$

M1

$$x^{\frac{1}{2}} = 2 \Rightarrow x = 2^2$$

$$x^{\frac{1}{2}} = k \ (k > 0), \text{ to } x = k^2.$$

PI by correct value of x if no error seen

m1

$M(4, 8)$

SC If M0 award B1 for (4, 8)

A1

3

(ii) Eqn of normal at M is $x = 4$

Ft on $x = c$'s x_M

B1F

1

(c) (i) When $x = \frac{9}{4}, \frac{dy}{dx} = 6 - 3 \times \frac{3}{2} = \frac{3}{2}$

Attempt to find $\frac{dy}{dx}$ when $x = \frac{9}{4}$

M1

Gradient of normal at $P = -\frac{2}{3}$
 $m \times m' = -1$ used

m1

Eqn of normal: $y - \frac{27}{4} = -\frac{2}{3}\left(x - \frac{9}{4}\right)$
 ACF eg $y = -\frac{2}{3}x + \frac{33}{4}$

A1

$12y - 81 = -8x + 18 \Rightarrow 8x + 12y = 99$
Coeffs and constant must now be positive integers, but accept different order
 eg $12y + 8x = 99$

A1

4

(ii) $8(4) + 12y = 99$
*Solving c's answer (b)(ii), (must be in form $x = \text{positive const}$), with c's answer (c)(i).
 PI by correct earlier work and correct coordinates for R.*

M1

$R\left(4, \frac{67}{12}\right)$

Accept 5.58 or better as equivalent to $\frac{67}{12}$

A1

2

[13]

Q6.

(a) (i) $y = xe^{2x}$

$\left(\frac{dy}{dx}\right) = 2xe^{2x} + e^{2x}$

$kxe^{2x} + le^{2x}$ where k and l are 1s or 2s

M1

A1

$$\left. \begin{matrix} k = 2 \\ l = 1 \end{matrix} \right\} \text{Independent of each other}$$

A1

$$(= e^{2x} (2x + 1))$$

ISW

3

(ii) $x = 1 \Rightarrow \frac{dy}{dx} = 3e^2$

correct substitution of $x = 1$ into their $\frac{dy}{dx}$ but must have earned M1 in part (i)

M1

tangent: $y - e^2 = 3e^2(x - 1)$ OE

CSO (no ISW), must have scored first 4 marks

common correct answer: $y = 3e^2 x - 2e^2$

A1

2

(b) $y = \frac{2 \sin 3x}{1 + \cos 3x}$

$$\left(\frac{dy}{dx} = \right) \frac{(1 + \cos 3x)6 \cos 3x - 2 \sin 3x(-3 \sin 3x)}{(1 + \cos 3x)^2}$$

$$\pm \frac{p(1 + \cos 3x) \cos 3x \pm q \sin 3x (\sin 3x)}{(1 + \cos 3x)^2}$$

where p and q are rational numbers

condone poor use/omission of brackets

PI by further working

M1

$$= \frac{6 \cos 3x + 6 \cos^2 3x + 6 \sin^2 3x}{(1 + \cos 3x)^2}$$

this line must be seen in this form (ie in terms of $\cos^2 3x$ and $\sin^2 3x$), but allow $\sin^2 3x$ replaced by $1 - \cos^2 3x$

condone denominator correctly expanded

A1

correct use of $k \sin^2 3x + k \cos^2 3x = k$
 or $k \sin^2 3x = k(1 - \cos^2 3x)$

m1

$$= \frac{6 \cos 3x + 6}{(1 + \cos 3x)^2}$$

$$= \frac{6}{1 + \cos 3x}$$

CSO

A1

4

note: if degrees used then no marks in (a) and (c)

[9]

Q7.

(a) (i) $\left(\frac{dx}{d\theta}\right) = -6 \sin 2\theta, \left(\frac{dy}{d\theta}\right) = -2 \sin \theta$
 $\left(\frac{dx}{d\theta}\right) = p \sin \theta \text{ or } r \sin \theta \cos \theta$
 $\left(\frac{dy}{d\theta}\right) = q \sin \theta$

M1

Both correct.

A1

$$\frac{dy}{dx} = \frac{-2 \sin \theta}{-6 \sin 2\theta}$$

$$\frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$$

Use chain rule $\frac{dy}{d\theta} \div \frac{dx}{d\theta}$; condone one slip

M1

$$= \frac{2 \sin \theta}{6 \times 2 \sin \theta \cos \theta} = \frac{1}{6 \cos \theta}$$

$k = 6$ must come from correct working seen **AG**

A1

(ii) $\theta = \frac{\pi}{3} \quad m_T = \frac{1}{3}$

ft on k $\left(\frac{1}{k \times \frac{1}{2}} \right)$
k need not be numerical

B1ft

$m_N = -3$

ft on m_T

B1ft

$(x, y) = \left(-\frac{3}{2}, 1 \right)$

B1

Normal $y - 1 = -3 \left(x + \frac{3}{2} \right)$
 CAO; any correct form, ISW.
 $2y + 6x + 7 = 0$

B1

(b) $\sin^2 x = \frac{1}{2} (1 - \cos 2x)$

$p + q \cos 2x$; Allow different letters for x or
 mixture eg θ even for A1 and the following A1ft

M1

A1

$\int p \, dx = px \quad \int q \cos 2x = \frac{1}{2} q \sin 2x$
 Both integrals correct;
 ft on p and q

A1ft

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sin^2 x dx = \left[\frac{x}{2} - \frac{1}{4} \sin 2x \right]$$

$$= \left(\frac{\pi}{8} - \frac{1}{4} \right) - \left(-\frac{\pi}{8} - \left(-\frac{1}{4} \right) \right)$$

Correct use of limits;

$$F\left(\frac{\pi}{4}\right) - F\left(-\frac{\pi}{4}\right) \text{ or } 2F\left(\frac{\pi}{4}\right)$$

$$F(x) = px + r \sin 2x \text{ and } \sin \frac{\pi}{2}, \sin\left(-\frac{\pi}{2}\right)$$

must be evaluated correctly for m1

m1

$$= \frac{\pi}{4} - \frac{1}{2}$$

CSO OE ISW

A1

5

Alternative

$$\int \sin^2 x dx = -\sin x \cos x - \int -\cos x \cos x dx$$

Use parts; condone sign slips

(M1)

$$= -\sin x \cos x + \int 1 - \sin^2 x dx$$

Use $\cos^2 x = 1 - \sin^2 x$

(m1)

$$2 \int \sin^2 x dx = -\sin x \cos x + x$$

(A1)

$$2 \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sin^2 x dx = G\left(\frac{\pi}{4}\right) - G\left(-\frac{\pi}{4}\right)$$

Correct use of limits

(m1)

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sin^2 x dx = \frac{\pi}{4} - \frac{1}{2}$$

(A1)

(5)

[13]

Q8.

(a) $(C =) \frac{2}{e}$ or $2e^{-1}$ or $2\left(\frac{1}{e}\right)$ or $2(e^{-1})$

*One of these answers only.
Not 0.736 but allow ISW.*

B1

1

(b) $\frac{d}{dx}(2y) = 2 \frac{dy}{dx}$

B1

$$\frac{d}{dx}(e^{2x}y^2) = 2e^{2x}y^2 + e^{2x}2y \frac{dy}{dx}$$

Product; 2 terms added, one with $\frac{dy}{dx}$;

M1

A1 for each term

A1

A1

$$\frac{d}{dx}(x^2 + C) = 2x$$

B1

$$\frac{dy}{dx} =$$

Solve their equation correctly for $\frac{dy}{dx}$

M1

$$\frac{x - e^{2x}y^2}{e^{2x}y + 1}$$

Condone factor of 2 in both numerator and denominator. ISW

A1

7

(c) Evaluate $\frac{dy}{dx}$ at $\left(1, \frac{1}{e}\right)$

Substitute $x = 1$ and $y = \frac{1}{e}$ into numerator of $\frac{dy}{dx}$;
allow one slip

M1

numerator = $1 - e^2 e^{-2} = 0 \Rightarrow$ stationary point

Conclusion required; must score full marks in part (b)

Allow $1 - 1 = 0$ or $2 - 2 = 0$

A1

2

[10]

Q9.

(a) (i) $\frac{dy}{dt} = \frac{4t^3}{8} - 4t + 4$

one term correct

another term correct

M1

A1

all correct (no + c etc) unsimplified

A1

3

(ii) $\frac{d^2y}{dt^2} = \frac{12t^2}{8} - 4$

ft one term "correct"

M1

correct unsimplified (penalise inclusion of +c once only in question)

A1

2

(b) $t = 2; \frac{dy}{dt} = 4 - 8 + 4$

Substitute $t = 2$ into their $\frac{dy}{dt}$

M1

$\frac{dy}{dt} = 0 \Rightarrow$ stationary value

CSO; shown = 0 plus statement

A1

$t = 2; \frac{d^2y}{dt^2} = 8 - 4 = 2$

Sub $t = 2$ into their $\frac{d^2y}{dt^2}$

M1

$\Rightarrow y$ has MINIMUM value
CSO

A1

4

(c) (i) $t = 1; \frac{dy}{dt} = \frac{1}{2} - 4 + 4$

Substitute $t = 1$ into their $\frac{dy}{dt}$

M1

$= \frac{1}{2}$

OE; CSO

NMS full marks if $\frac{dy}{dt}$ correct

A1

2

(ii) $\frac{dy}{dt} > 0 \Rightarrow$ (depth is) INCREASING

$\frac{dy}{dt}$
 allow decreasing if states that their
Reason must be given not just the word increasing or decreasing

E1ft

1

[12]

Q10.

(a) (i) $\frac{dy}{dx} = 24x - 19 - 6x^2$
2 terms correct

M1

all correct (no + c etc)

A1

when $x = 2, \frac{dy}{dx} = 48 - 19 - 24$

m1

$\Rightarrow$ gradient = 5
 CSO

A1

4

(ii) grad of normal = $-\frac{1}{5}$
ft their answer from (a)(i)

B1ft

$y + 6 = \left(\text{their} - \frac{1}{5} \right) x - 2$

or $y = \left(\text{their} - \frac{1}{5} \right) x + c$ and c evaluated

*ft grad of their normal using **correct** coordinates BUT must not be tangent*
condone omission of brackets

M1

using $x = 2$ and $y = -6$
 $x + 5y + 28 = 0$

CSO; condone all on one side in different order

A1

3

(b) (i) $\frac{12}{3}x^3 - \frac{19}{2}x^2 - \frac{2}{4}x^4$

one term correct

M1

another term correct

A1

all correct (ignore + c or limits)

A1

$= 32 - 38 - 8$

F 2 attempted

m1

$= -14$

CSO; withhold A1 if changed to +14 here

A1

5

(ii) Area $\Delta = \frac{1}{2} \times 2 \times 6 = 6$
condone -6

B1

Shaded region area = $14 - 6$

difference of $\pm \int \pm \Delta$

$= 8$

CSO

A1

3

[15]

Q11.

(a) $\frac{1}{x^3} = x^{-3}$

PI by its correct derivative

B1

$$\frac{dy}{dx} = -3x^{-4} + 48$$

A power decreased by 1; could be the +48 or the ft after B0

M1A1

3

(b) $-3x^{-4} + 48 = 0$

c's answer to (a) equated to 0

M1

$$x^{-4} = 16$$

*To $x^p = q$ but only ft on eqns of the form $ax^{2k} + 48 = 0$, where a and k are **negative integers***

A1F

$$x = \pm \frac{1}{2}$$

A1

Eqns of tangents: $y = 32$ and $y = -32$

Only ft if answer is of the form $y = \pm k$

A1F

4

(c) When $x = 1$, $\frac{dy}{dx} = -3 + 48 = 45$

Attempt to find value of $\frac{dy}{dx}$ at $x = 1$

M1

Gradient of normal at $(1, 49)$ is $-\frac{1}{45}$

Correct use of $m \times m' = -1$ with c's value of

$\frac{dy}{dx}$ when $x = 1$

m1

Normal at $(1, 49)$: $y - 49 = -\frac{1}{45}(x - 1)$

CSO. Apply ISW after ACF;

accept 49.02 or better in place of $49\frac{1}{45}$

A1

3

[10]

Q12.

(a) $y' = e^{-4x}(2x + 2) - 4e^{-4x}(x^2 + 2x - 2)$
 $y' = Ae^{-4x}(ax + b) \pm Be^{-4x}(x^2 + 2x - 2)$
 where A and B are non-zero constants

M1

All correct

A1

$$= e^{-4x}(2x + 2 - 4x^2 - 8x + 8)$$

$$\text{or } -4x^2e^{-4x} - 6xe^{-4x} + 10e^{-4x}$$

$$= 2e^{-4x}(5 - 3x - 2x^2)$$

AG; all correct with no errors,
 2nd line (OE) must be seen
 Condone incorrect order on final line

A1

or

$$y = x^2e^{-4x} + 2xe^{-4x} - 2e^{-4x}$$

$$y' = -4x^2e^{-4x} + 2xe^{-4x} + 2x \cdot -4e^{-4x} + 2e^{-4x} + 8e^{-4x}$$

$$Ax^2e^{-4x} + Bxe^{-4x} + Cxe^{-4x} + De^{-4x} + Ee^{-4x}$$

(M1)

All correct

(A1)

$$= -4x^2e^{-4x} - 6xe^{-4x} + 10e^{-4x}$$

$$= 2e^{-4x}(5 - 3x - 2x^2)$$

AG; all correct with no errors,
 3rd line (OE) must be seen

(A1)

3

(b) $-(2x + 5)(x - 1) (= 0)$

OE Attempt at factorisation
 $(\pm 2x \pm 5)(\pm x \pm 1)$
or formula with at most one error

M1

$$x = \frac{-5}{2}, 1$$

Both correct and no errors
SC $x = 1$ only scores M1A0

A1

$$x = 1, y = e^{-4}$$

For $y = ae^b$ attempted

m1

Either correct, follow through only from
incorrect sign for x

A1F

$$x = -\frac{5}{2}, y = e^{10}\left(-\frac{3}{4}\right)$$

CSO 2 solutions only
Note: withhold final mark for extra solutions
Note: approximate values only for y can
score m1 only

A1

EXAM PAPERS PRACTICE⁵

[8]

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Q13.

$$x^2 + xy = e^y$$

$$2x + y + x \frac{dy}{dx} = e^y \frac{dy}{dx}$$

$$2x$$

B1

Use product rule

M1A1

RHS

B1

$$(-1, 0) \quad \frac{dy}{dx} = -1$$

CSO

A1

[5]

Q14.

(a) (i) $\sin 2\theta = 2 \sin \theta \cos \theta$

B1

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

OE condone use of x etc, but variable must be consistent

B1

2

(ii) $\sin \theta = \frac{4}{5} \Rightarrow \sin 2\theta = 2 \times \frac{4}{5} \times \frac{3}{5} = \frac{24}{25}$

AG

Use of 106.26° B0

B1

or

$$2 \times \sin \left(\cos^{-1} \frac{3}{5} \right) \times \frac{3}{5}$$

$$\cos 2\theta = \frac{9}{25} - \frac{16}{25} = -\frac{7}{25}$$

-0.28

B1

2

(b) (i) $\frac{dx}{d\theta} = 6 \cos 2\theta \quad , \quad \frac{dy}{d\theta} = -8 \sin 2\theta$

Attempt both derivatives. ie $p \cos 2\theta$

M1

Both correct.

$q \sin 2\theta$

A1

$$\frac{dy}{dx} = -\frac{4 \sin 2\theta}{3 \cos 2\theta} \quad \text{ISW}$$

CSO OE

A1

3

(ii) $P\left(\frac{72}{25}, -\frac{28}{25}\right)$
 $(2.88, -1.12)$

B1F

Gradient $= -\frac{4}{3} \times -\frac{24}{7}$

Their $\frac{q \sin 2\theta}{p \cos 2\theta}$ or $\frac{p \cos 2\theta}{q \sin 2\theta}$

must be working with rational numbers

M1

Tangent $y + \frac{28}{25} = \frac{32}{7} \left(x - \frac{72}{25} \right)$ ISW

Any correct form.

$7y = 32x - 100$

Fractions in simplest form

Equation required

A1

3

[10]

Q15.

(a) (i) $\frac{x^5}{5} - \frac{8}{2}x^2 + 9x$

One term correct

M1

Another term correct

A1

All correct (may have + c)

A1

$\frac{32}{5} - 16 + 18$

F(2) attempted

m1

$$= 8\frac{2}{5}$$

$$\frac{42}{5}, 8.4$$

A1

5

- (ii) Shaded area = $18 - \text{'their integral'}$
PI by 18 – (a)(i) NMS

M1

$$= 9\frac{3}{5}$$

$$\frac{48}{5}, 9.6 \text{ NMS full marks}$$

A1

2

(b) (i) $\frac{dy}{dx} = 4x^3 - 8$

One term correct

M1

All correct (no + c etc)

A1

$$x = 1 \Rightarrow \frac{dy}{dx} = 4 - 8$$

sub $x = 1$ into their $\frac{dy}{dx}$

m1

(Gradient of curve) = -4
No ISW

A1 cso

4

- (ii) $y - 2 = -4(x - 1); y = -4x + c, c = 6$
any correct form; FT their answer from (b)(i) but must use $x = 1$ and $y = 2$

B1ft

1

Q16.

- (a) (i) S.A. = $4xy + 5xy + 3xy + 6x^2$ OE
Condone one slip or omission

M1

$$= 12xy + 12x^2$$

A1

$$144 = 12xy + 12x^2$$

Must see this line

$$\Rightarrow xy + x^2 = 12$$

AG

A1cso

3

(ii) (Volume =) $\frac{1}{2} \times 3x \times 4x \times y$ OE

M1

$$= 6x^2 \times \frac{(12 - x^2)}{x}$$

$$\frac{(12 - x^2)}{x}$$

*Must see ($y = \frac{(12 - x^2)}{x}$ or $xy = 12 - x^2$)
 for A1*

$$(V =) 72x - 6x^3$$

*AG must be convinced not working back
 from answer*

A1

2

(b) (i) $\frac{dV}{dx} = 72 - 18x^2$

One term correct

M1

All correct (no + c etc)

A1

2

(ii) $x = 2 \Rightarrow \frac{dV}{dx} = 72 - 18 \times 2^2$
 $\Rightarrow \frac{dV}{dx} = 72 - 72 = 0$

Substitute $x = 2$ into their $\frac{dV}{dx}$

M1

$\Rightarrow$ stationary (value when $x = 2$)
Shown = 0 plus statement
Statement may appear first

A1

2

(c) $\frac{d^2V}{dx^2} = -36x$

FT their $\frac{dV}{dx}$

B1ft

$\frac{d^2V}{dx^2} = -72$ or when $x = 2 \Rightarrow \frac{d^2V}{dx^2} < 0 \Rightarrow$ maximum

FT their $\frac{d^2V}{dx^2}$ value when $x = 2$
 with appropriate conclusion

E1ft

2

[11]

Q17.

(a) $\sqrt{x} = x^{\frac{1}{2}}$

PI

B1

$$\frac{x^3 + \sqrt{x}}{x} = \frac{x^3}{x} + \frac{\sqrt{x}}{x} = x^2 + x^{-\frac{1}{2}}$$

Accept $p = 2$; $q = -\frac{1}{2}$

B1;B1

3

(b) (i) $\frac{dy}{dx} = 2x - \frac{1}{2}x^{-\frac{3}{2}}$

Reduces both powers by 1

M1

ACF

A1

2

(ii) When $x = 1$, $y = 2$

PI if not stated explicitly eg the '2' may appear in the correct posn. in later eqn.

B1

When $x = 1$, $\frac{dy}{dx} = 2 - \frac{1}{2} = \frac{3}{2}$

Attempt to find $\frac{dy}{dx}$ When $x = 1$ PI

M1

Gradient of normal = $-\frac{2}{3}$

*-1/ (c's value of dy/dx when $x = 1$)
either stated as the gradient of the
normal or used as the gradient in the
equation of the normal*

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m1

Equation of normal: $y - 2 = -\frac{2}{3}(x - 1)$

*Only ft on c's $\frac{dy}{dx}$ in part (b)(i)
ACftF*

A1F

4

(c) (i) $\frac{d^2y}{dx^2} = 2 + \frac{3}{4}x^{-\frac{5}{2}}$

Reduces both powers by 1.

M1

Ft on (b)(i) provided at least one power to be differentiated is both negative and fractional

A1F

2

(ii) (Since $x > 0$,) $\frac{d^2 y}{dx^2} > 0$

For a maximum point $\frac{d^2 y}{dx^2}$ is **not** positive so C has no maximum points

*E1 for attempt to find the sign of $\frac{d^2 y}{dx^2}$;
either in general terms or at the pt(s)
where c 's $dy/dx = 0$ or the remaining E mark*

*a correct justification for why $\frac{d^2 y}{dx^2} > 0$ and also
a full correct concluding statement must be made.*

E2,1,0

2

[13]

Q18.

(a) $y = \frac{\ln x}{x}$

(when) $y = 0$ $x = 1$ or $(1, 0)$

Both coordinates must be stated, not 1 simply shown on diagram

B1

1

(b) $\left(\frac{dy}{dx}\right) = \frac{x \times \frac{1}{x} - \ln x}{x^2}$

Quotient/product rule $\frac{\pm \frac{x}{x} \pm \ln x}{x^2}$

M1

$= \frac{1 - \ln x}{x^2}$ or $x^{-2} - x^{-2} \ln x$

OE must simplify $\frac{x}{x}$

A1

At B, $\frac{1 - \ln x}{x^2} = 0$

Putting their $\frac{dy}{dx} = 0$ or numerator = 0

m1

$x = e$

CSO condone $x = e^1$

A1

$y = \frac{1}{e} \text{ or } e^{-1}$

CSO must simplify $\ln e$

A1

5

(c) Gradient at $x = e^3$
 $= \frac{1 - \ln e^3}{(e^3)^2}$

Substituting $x = e^3$ into their $\frac{dy}{dx}$ (condone 1 slip) but must have scored M1 in (b)

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$= \frac{-2}{e^6} \text{ or } -2e^{-6}$

PI

A1

Gradient of normal $= \frac{1}{2} e^6$
CSO simplified to this

A1

3

[9]

Q19.

(a) $\frac{dx}{dt} = -3 \quad \frac{dy}{dt} = 6t^2$

Both derivatives correct; PI

B1

$$\frac{dy}{dx} = -\frac{6t^2}{3}$$

Correct use of chain rule

M1

$$= -2t^2$$

CSO

A1

3

(b) $t = 1 \quad m_T = -2 \quad m_N = \frac{1}{2}$

Substitute $t = 1 \quad m_N = -\frac{1}{m_T}$

M1

F on gradient; $m_T \neq \pm 1$

A1F

Attempt at equation of normal using $(x, y) = (-2, 3)$

Condone one error

M1

Normal has equation $y - 3 = \frac{1}{2}(x + 2)$

CSO; ACF

A1

4

(c) $t = \frac{1-x}{3} \quad \text{or} \quad t = \sqrt[3]{\frac{y-1}{2}}$

Correct expression for t in terms of x or y

M1

$$y = 1 + 2\left(\frac{1-x}{3}\right)^3$$

ACF

A1

2

[9]

Q20.

(a) $x^3 + \cos \pi = 7 \Rightarrow x^3 - 1 = 7$
 Or $x = \sqrt[3]{7 - \cos \pi}$

M1

$$x = 2$$

CSO

A1

2

(b) $\frac{d}{dx}(x^3 y) = 3x^2 y + x^3 \frac{dy}{dx}$
 2 terms added, one with $\frac{dy}{dx}$

M1A1

$$\frac{d}{dx}(\cos \pi y) = -\pi \sin(\pi y) \frac{dy}{dx}$$

B1

At (2,1) $3 \times 4 + 8 \frac{dy}{dx} - \pi \sin \pi \frac{dy}{dx} = 0$

*Substitute candidate's x from (a) and
 y = 1 with 0 on RHS and both derivatives
 attempted and no extra derivatives*

M1

$$\frac{dy}{dx} = -\frac{3}{2}$$

CSO; OE

A1

5

[7]

Q21.

(a) (i) $\frac{dx}{dt} = 2t^3 - 40t + 66$

one term correct

M1

another term correct

A1

all correct unsimplified (no + c etc)

A1

3

(ii) $\frac{d^2x}{dt^2} = 6t^2 - 40$

ft one term correct

M1

ft all "correct", 2 terms equivalent

A1ft

2

(b) $\frac{dx}{dt} = 54 - 120 + 66$

substitute $t = 3$ into their $\frac{dx}{dt}$

M1

$= 0 \Rightarrow$ stationary value

CSO

shown = 0 (54 or 2×27 seen) and statement

A1

Substitute $t = 3$ into $\frac{d^2x}{dt^2} (= 14)$

M1

$\frac{d^2x}{dt^2} > 0 \Rightarrow$ minimum value

CSO; all values (if stated) must be correct

A1

4

- (c) Substitute $t = 1$ into their $\frac{dx}{dt}$
must be their $\frac{dx}{dt}$ NOT $\frac{d^2x}{dt^2}$ etc

M1

$$\frac{dx}{dt} = 28$$

ft their $\frac{dx}{dt}$ when $t = 1$

A1ft

2

- (d) Substitute $t = 2$ into their $\frac{dx}{dt}$
must be their $\frac{dx}{dt}$ NOT $\frac{d^2x}{dt^2}$ or x
Interpreting their value of $\frac{dx}{dt}$

M1

$$= 16 - 80 + 66 = 2 \quad (> 0)$$

$\Rightarrow$ increasing when $t = 2$

Allow decreasing if their $\frac{dx}{dt} < 0$

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2

[13]

Q22.

- (a) (i) $p(2) = 8 + 2 - 10$
Must find $p(2)$ NOT long division

M1

$\Rightarrow p(2) = 0 \Rightarrow (x - 2)$ is factor
Shown = 0 plus a statement

A1

2

- (ii) Attempt at long division (generous)

Obtaining a quotient $x^2 + cx + d$
or equating coefficients (full method)

M1

$$p(x) = (x - 2)(x^2 + 2x + 5)$$

$a = 2, b = 5$ by inspection B1, B1

A1

2

(b) (i) $\frac{dy}{dx} = 3x^2 + 1$

One term correct

M1

All correct – no + c etc

A1

When $x = 2 \quad \frac{dy}{dx} = 3 \times 4 + 1$

Sub $x = 2$ into their $\frac{dy}{dx}$

m1

Therefore gradient at Q is 13
CSO

A1

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4

(ii) $y = 13(x - 2)$

Tangent (NOT normal) attempted
ft their gradient answer from (b)(i)

M1

CSO; correct in any form

A1

2

(iii) $\int \dots dx = \frac{x^4}{4} + \frac{x^2}{2} - 10x (+ c)$

one term correct

M1

second term correct

A1

all correct (condone no +c)

A1

3

(iv) $[4 + 2 - 20] - [0] = -14$

F(2) attempted and possibly F(0)

Must have earned M1 in (b)(iii)

M1

Area of shaded region = 14

CSO; separate statement following correct evaluation of limits

A1

2

[15]

Q23.

(a) (i) $\frac{dy}{dx} = 3x^{\frac{1}{2}}$
 $kx^{\frac{1}{2}}$ with or without + c

M1

= 6 {when $x = 4$ }

Must be 6 and seen in (a)(i)

6 + c is A0

A1cao

2

(ii) y -coordinate of A = $2 \times 4^{\frac{3}{2}} (= 16)$

Substitute $x = 4$ in $y = 2x^{\frac{3}{2}}$

M1

$6 \times m' = -1$

$m_1 \times m_2 = -1$ OE used with c's value of

$\frac{dy}{dx}$ when $x = 4$. PI

M1

$$y - 16 = m(x - 4)$$

dep on 1st M1 in (a)(ii)
m must be numerical

m1

$$y - 16 = -\frac{1}{6}(x - 4)$$

ACF

A1

4

(b) (i)

$$\int 8x^{\frac{1}{2}} dx = \frac{8}{\frac{3}{2}} x^{\frac{1}{2}+1} \{+c\}$$

Index raised by 1

M1

$$= \frac{16}{3} x^{\frac{3}{2}} \{+c\}$$

Condone missing '+ c'
Coefficient must be simplified

A1

2

(ii)

$$\int 2x^{\frac{3}{2}} dx = \frac{2}{\frac{5}{2}} x^{\frac{5}{2}} \{+c\} \quad \{= \frac{4}{5} x^{\frac{5}{2}} \{+c\}\}$$

Can award for unsimplified form

B1

$$\int_0^4 8x^{\frac{1}{2}} dx - \int_0^4 2x^{\frac{3}{2}} dx$$

Ignore limits here

M1

$$= \frac{16}{3} (4)^{\frac{3}{2}} - 0 - \left[\frac{4}{5} (4)^{\frac{5}{2}} - 0 \right]$$

F(4) – F(0) used in either; {F(0) = 0 PI}
Cand. must be using F(x) as a result of
his/her integration in (b)(i) or in the (b)(ii)
B1 line above

M1

$$= \frac{256}{15}$$

Accept any value from 17.04 to 17.1
inclusive in place of 256/15

A1

4

(c) Translation

Accept 'translat...' as equivalent
[T or Tr is NOT sufficient]

B1

$$\begin{bmatrix} -3 \\ 0 \end{bmatrix}$$

Accept equivalent in words provided
linked to 'translation/move/shift'
(B0B0 if >1 transformation)

B1

2

[14]

Q24.

(a) $(1 + 2x)^4 = 1 + 4(2x) + 6(2x)^2 + 4(2x)^3 + (2x)^4$

M1

$$= 1 + 8x$$

A1

$$+ 24x^2$$

A1

$$+ 32x^3 \{+ 16x^4\}$$

A1

(1), 4, 6, 4, (1) OE unsimplified with
correct powers of x
Algebraic multiplication must be a full
method

Accept $a = 8$ provided 1st term is 1

$$b = 24$$

$$c = 32$$

4

(b) $(1 - 2x)^4 = 1 - 8x + 24x^2 - 32x^3 \{+ 16x^4\}$

Replace x by $-x$ even in M1 line of (a) PI

M1

ft c's non zero values for a , b and c

A1Ft

$$\begin{aligned} & (1 + 2x)^4 + (1 - 2x)^4 \\ &= 1 + 8x + 24x^2 + 32x^3 + 16x^4 + 1 - 8x + 24x^2 - 32x^3 + 16x^4 \\ &= 2 + 48x^2 + 32x^4 \end{aligned}$$

AG Be convinced

A1 cso

3

(c) $\frac{dy}{dx} = 96x + 128x^3$

A correct power of x OE

M1

For st. pt $96x + 128x^3 = 0$

A1

$$32x(3 + 4x^2) = 0$$

Since $3 + 4x^2 > 0$ there is only one stationary point

Any valid explanation of curve having just one stationary point

E1

The coordinates of the stationary point are (0, 2)

*(0, 2) as the **only** stationary point*

B1

4

[11]

Q25.

(a) $y = e^{2x}(x^2 - 4x - 2)$

$$\frac{dy}{dx} = e^{2x}(2x - 4)$$

Product rule; allow 1 slip

M1

$$+ (x^2 - 4x - 2)2e^{2x}$$

A1

$$\frac{dy}{dx} = e^{2x}(2x - 4 + 2x^2 - 8x - 4)$$

Factorising $e^{2x}(2x^2 - 6x - 8)$

M1

$$e^{2x}(2x^2 - 6x - 8)$$

$$\text{Or } x^2 - 3x - 4 = 0$$

A1

$$e^{2x} \neq 0$$

$$(x - 4)(x + 1) = 0$$

*Solving 3 term quadratic
Dependent on both M marks*

m1

$$x = 4, -1$$

And no extras eg $x = 0$

A1

6

(b) (i) $\frac{d^2y}{dx^2} = e^{2x} \cdot 2 + (2x - 4)2e^{2x}$

*Product rule from their $\frac{dy}{dx}$ in form
 e^{2x} (quadratic) $e^{2x}(4x^2 - 8x - 22)$*

M1

$$+ (x^2 - 4x - 2)4e^{2x} + 2e^{2x}(2x - 4)$$

A1

OR

$$\frac{d^2y}{dx^2} = e^{2x}(4x - 6) + (2x^2 - 6x - 8)2e^{2x}$$

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M1A1

2

(ii) $x = 4 : y'' = e^8 (10) > 0 \therefore \text{MIN}$

*Their 2 x's in their $\frac{d^2y}{dx^2}$
only of form e^{2x} (quadratic)*

M1

$$x = -1 : y'' = e^{-2} (-10) < 0 \therefore \text{MAX}$$

*CSO Both correct
Allow values either side of y or y'*

A1

Q26.

(a) $x^2 \frac{dy}{dx} + 2xy$

Product rule used. Allow 1 error

M1A1

$+ 3y^2 \frac{dy}{dx}$

Chain rule

B1

$= 2$

RHS and equation with no spurious

$\frac{dy}{dx}$ *unless recovered.*

B1

$(2, 1), \quad 4 \frac{dy}{dx} + 4 + 3 \frac{dy}{dx} = 2$

Substitute (2, 1)

M1

$\frac{dy}{dx} = -\frac{2}{7}$

CSO

A1

6

(b) $\frac{dy}{dx} = 0 \Rightarrow$

Derivative = 0 used

M1

$xy = 1$

OE

A1

$x^2 \times \frac{1}{x} + \frac{1}{x^3} = 2x + 1$

Use $xy = k$ to eliminate y on LHS

m1

$$\frac{1}{x^3} = x + 1$$

Answer given; CSO

A1

4

[10]

Q27.

(a) $\frac{dy}{dx} = 5x^4 + 40x$

One of these powers correct

M1

One of these terms correct

A1

All correct (no + c etc)

A1

3

(b) $x = -2 \quad \frac{dy}{dx} = 5 \times (-2)^4 + (40 \times -2)$

Substitute $x = -2$ into their $\frac{dy}{dx}$

M1

$$\frac{dy}{dx} = 5 \times 16 + (40 \times -2) = 0 \Rightarrow P \text{ is stationary point}$$

*CSO Shown = 0 plus statement
eg "st pt", "as required", "grad = 0" etc*

A1

Or *their* $\frac{dy}{dx} = 0 \Rightarrow x^* = k$

(M1)

$$x^3 = -8 \Rightarrow x = -2$$

CSO $x = 0$ need not be considered

(A1)

2

(c) (i) $\frac{d^2y}{dx^2} = 20x^3 + 40$

Correct ft their $\frac{dy}{dx}$

B1

$= 20 \times (-2)^3 + 40$

Subst $x = -2$ into their second derivative

M1

$(= -160 + 40) = -120$

CSO

A1

3

- (ii) Maximum (value)
 their c(i) answer must be < 0
 Other valid methods acceptable provided
 "maximum" is the conclusion
*Accept minimum if their c(i) answer > 0
 and correctly interpreted
 Parts (i) and (ii) may be combined by
 candidate but -120 must be seen to award
 A1 in part (c)(i)*

E1ft

1

(d) (When $x = 1$) $y = 13$

B1

When $x = 1$, $\frac{dy}{dx} = 5 + 40$

Sub $x = 1$ into their $\frac{dy}{dx}$

M1

$y = (\text{their } 45)x + k$ OE

ft their $\frac{dy}{dx}$

m1

Tangent has equation $y - 13 = 45(x - 1)$

CSO OE $y = 45x + c, \quad c = -32$

A1

4

[13]

Q28.

(a) $\frac{dy}{dx} = \frac{45}{2}x^{\frac{1}{2}} - \frac{5}{2}x^{\frac{3}{2}}$

One power correctly obtained

M1

A1 for each term on the RHS coeffs simplified

A2,1,0

3

(b) $\frac{45}{2}x^{\frac{1}{2}} - \frac{5}{2}x^{\frac{3}{2}} = 0$

can't's (a) = 0

M1

$\frac{5}{2}x^{\frac{1}{2}}(9 - x) = 0$

Must be solving eqn of form $ax^m + bx^n = 0$, m and n non-zero, with at least one of m and n non-integer and reaching a stage from which the non-zero value of x can be stated PI. Must deal with powers of x correctly and any squaring of kx^p terms or expressions must be correct.

m1

At M, $x = 9$

A1

$y_M = 162$

M1 must be scored, else 0/4

A1

4

(c) At $P(1, 14), \frac{dy}{dx} = \frac{45}{2} - \frac{5}{2} = 20$

Attempt to find $y'(1)$

M1

Tangent at $P: y - 14 = m(x - 1)$

$m = \text{cand's value of } y'(1)$

m1

$y - 14 = 20x - 20; y = 20x - 6$

CSO; AG

A1

3

(d) Tangent at $M: y = 162$

ft $y = \text{cand's } y_M$

B1F

At $R, 162 = 20x - 6; x = 8.4$

Solving cand's numerical $y_M = 20x - 6$ to find a value for x

M1

Distance $RM = |x_M - x_R| = 9 - 8.4 = 0.6$

ft on coordinates of M

A1F

3

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[13]

Q29.

(a) (i) $f(x) = \frac{\cos x}{2x+1} - \frac{1}{2}$

OE

$f(0) = \frac{1}{2}; f\left(\frac{\pi}{2}\right) = -\frac{1}{2}$

$x = 0 \quad LHS = 1, \quad x = \frac{\pi}{2} \quad LHS = 0$

M1

Change of sign $0 < \alpha < \frac{\pi}{2}$

Either side of $\frac{1}{2}, \therefore 0 < \alpha < \frac{\pi}{2}$

A1

2

(ii) $\frac{\cos x}{2x+1} = \frac{1}{2}$
 $\left. \begin{array}{l} 2\cos x = 2x+1 \\ 2\cos x - 1 = 2x \end{array} \right\} \text{or, } \cos x = x + \frac{1}{2}$

Either line

$$x = \cos x - \frac{1}{2}$$

$\cos x - \frac{1}{2} = x$
 AG; or
All correct with no errors

B1

1

(iii) $x_1 = 0$
 $x_2 = 05$

Attempt at iteration
(allow $x_2 = -0.5, x_3 = 0.38, 0.4$)

M1

$$x_3 = 0.378$$

CAO

A1

2

(b) (i) $\frac{dy}{dx} = \frac{(2x+1)(-\sin x) - \cos x \times 2}{(2x+1)^2}$

Attempt at quotient rule:

$$\frac{\pm (2x+1)\sin x \pm 2\cos x}{(2x+1)^2}$$

M1

Either term correct

A1

All correct ISW

A1

3

(ii) $x = 0$
 $\frac{dy}{dx} = -2$

Correctly subst. $x = 0$ into their $\frac{dy}{dx}$

m1

$\therefore$ Gradient of normal = $\frac{1}{2}$
 CSO

A1

2

[10]

Q30.

(a) $\frac{dx}{dt} = \frac{1}{t^2} \quad \frac{dy}{dt} = 1 - \frac{1}{2t^2}$

B1B1

$$\frac{dy}{dx} = \frac{1 - \frac{1}{2t^2}}{-\frac{1}{t^2}} \left(= \frac{2t^2 - 1}{-2} \right)$$

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Their $\frac{\frac{dy}{dt}}{\frac{dx}{dt}}$; condone 1 slip

M1

CSO; ISW

A1

Alternative

$$y = \frac{1}{x} + \frac{x}{2}$$

(B1)

$$\frac{dy}{dx} = -\frac{1}{x^2} + \frac{1}{2}$$

(B1)

Substitute $x = \frac{1}{t}$

(M1)

$$\frac{dy}{dx} = -t^2 + \frac{1}{2}$$

CSO

(A1)

4

(b) $t = 1 \quad \frac{dy}{dx} = -\frac{1}{2}$

Substitute $t = 1$ in $\frac{f(t)}{g(t)} \neq k$

M1

$$m_T = -\frac{1}{2} \Rightarrow m_N = 2$$

F on $m_T \neq 0$; if in $t \rightarrow$ numerical later

B1F

$$(x, y) = (1, \frac{3}{2})$$

$$Pl \frac{3}{2} = m(x1) + c$$

B1

$$(y - \frac{3}{2}) = 2(x - 1) \text{ or } y = 2x + c, c = -\frac{1}{2}$$

ISW, CSO (a) and (b) all correct

A1

4

(c) $y = \frac{1}{\frac{1}{t}} + \frac{1}{2} \times \frac{1}{t}$

Attempt to use $t = \frac{1}{x}$ to eliminate t
 t , or equivalent

M1

$$= \frac{1}{x} + \frac{x}{2}$$

A1

$$2xy = 2 + x^2 \Rightarrow x^2 - 2xy + 2 = 0$$

Correct algebra to AG with $k = 2$

allow $k = 2$ stated $k = 2$,

no working or from $\left(1, \frac{1}{3}\right)$: 0/3

A1

Alternative

or

$$\left(\frac{1}{t}\right)^2 - 2\left(\frac{1}{t}\right)\left(t + \frac{1}{2t}\right) \Bigg|_{xy} = \frac{1}{t}\left(t + \frac{1}{2t}\right)$$

$$= -2 \qquad \qquad \qquad = 1 + \frac{x^2}{2}$$

Substitute and multiply out

(M1)

Eliminate t

(A1)

$$\Rightarrow x^2 - 2xy + 2 = 0$$

Conclusion, $k = 2$

(A1)

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3

[11]

Q31.

$$8x + 2y \frac{dy}{dx} = 3y + 3x \frac{dy}{dx}$$

$$8x \text{ and } 4 \rightarrow 0$$

B1

$$2y \frac{dy}{dx}$$

B1

$$3y + 3x \frac{dy}{dx}$$

Two terms with one $\frac{dy}{dx}$

M1A1

at (1, 3) (gradient) $\frac{dy}{dx} = \frac{1}{3}$
CSO

A1

[5]

Q32.

(a) $\frac{dy}{dx} = 4x^3 - 32$

Reduce one power by 1

M1

One term correct

A1

All correct (no + c etc)

A1

3

(b) Stationary point $\Rightarrow \frac{dy}{dx} = 0$

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$$\Rightarrow x^3 = 8$$

$$x^n = k \text{ following from their } \frac{dy}{dx}$$

A1ft

$$\Rightarrow x = 2$$

CSO

A1

3

(c) (i) $\frac{d^2y}{dx^2} = 12x^2$

FT their $\frac{dy}{dx}$

B1ft

1

(ii) When $x = 2$, $\frac{d^2y}{dx^2}$ considered

Or complete test with $2 \pm \varepsilon$ using $\frac{dy}{dx}$

M1

$\Rightarrow$ minimum point

E1ft

2

(d) Putting $x = 0$ into their $\frac{dy}{dx}$ ($= -32$)

M1

$\frac{dy}{dx} < 0 \Rightarrow$ decreasing

Allow "increasing" if their $\frac{dy}{dx} > 0$

A1ft

2

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[11]

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Q33.

(a) (i) $p(-1) = -1 + 7 - 6$
Finding $p(-1)$

M1

$= 0$ therefore $x + 1$ is a factor
Shown to $= 0$ **plus statement**

A1

2

(ii) $p(x) = (x + 1)(x^2 - x - 6)$
Long division/inspection (2 terms correct)

M1

Quadratic factor correct

A1

$$p(x) = (x + 1)(x + 2)(x - 3)$$

May earn M1, A1 For correct second factor
then A1 For $(x + 1)(x + 2)(x - 3)$

A1

3

(b) (i) $A(-2, 0)$

Condone $x = -2$

B1

1

(ii) $\frac{x^4}{4} - \frac{7x^2}{2} - 6x (+c)$

One term correct

M1

Another term correct

A1

(may have + c or not)

All correct unsimplified

A1

$$\left[\frac{81}{4} - \frac{63}{2} - 18 \right] - \left[\frac{1}{4} - \frac{7}{2} + 6 \right]$$

$F(3) - F(-1)$ attempted in correct order

m1

$$= -32$$

CSO; OE

A1

5

(iii) Area of shaded region = 32

FT their (b)(ii) but positive value needed

B1ft

1

(iv) $\frac{dy}{dx} = 3x^2 - 7$

One term correct

M1

All correct (no + c etc)

A1

When $x = -1$, gradient = -4

CSO

A1

3

(v) Gradient of normal = $\frac{1}{4}$

B1ft

$y = \text{"their gradient"} (x \pm 1)$

*Must be finding **normal**, not tangent*

M1

$$y = \frac{1}{4}(x + 1)$$

CSO; any correct form eg $4y - x = 1$

A1

3

[18]

Q34.

(a) (i) $\frac{dy}{dx} = 4 \times \frac{1}{2} x^{-\frac{1}{2}} - \frac{3}{2} x^{\frac{1}{2}} = 2x^{-\frac{1}{2}} - \frac{3}{2} x^{\frac{1}{2}}$

A power decreased by 1

M1

A1 For each correct term

A1 A1

3

(ii) At $P(4, 0)$, $\frac{dy}{dx} = \frac{2}{\sqrt{4}} - \frac{3}{2} \times 2$

Attempts $\frac{dy}{dx}$ when $x = 4$

M1

$$= 1 - 3 = -2$$

AG

A1

2

(iii) Gradient of normal = $\frac{1}{2}$

Use of or stating $m \times m' = -1$

M1

Equation of normal is $y - 0 = m [x - 4]$

m numerical; can be awarded even if $m = -2$

M1

$$y - 0 = \frac{1}{2}(x - 4) \Rightarrow 2y = x - 4$$

ACF of the equation

A1

3

(iv) At Q, $x = 0$, $2y = 0 - 4$
PI

M1

$$y_e = -2$$

Area of triangle $OPQ = 0.5 \times 4 \times |y_e|$

Ft on a linear equation for normal provided y_e is negative and prev A1 is lost

A1F

$$= 4$$

Ft on c's negative y_e

B1F

3

(v) $2x^{-\frac{1}{2}} - \frac{3}{2}x^{\frac{1}{2}} = 0 \Rightarrow 2x^{-\frac{1}{2}} = \frac{3}{2}x^{\frac{1}{2}}$

Puts c's $\frac{dy}{dx} = 0$ and a 1st step in attempt to solve.

M1

$$2 = \frac{3}{2}x \quad \Rightarrow x = \frac{4}{3}$$

Valid method to $ax = b$

m1

Condone 1.3 or better

A1

3

(b) (i)

$$\int \left(4x^{\frac{1}{2}} - x^{\frac{3}{2}} \right) dx = 4 \frac{x^{\frac{3}{2}}}{1.5} - \frac{x^{\frac{5}{2}}}{2.5} \{+c\}$$

$$= \frac{8}{3} x^{\frac{3}{2}} - \frac{2}{5} x^{\frac{5}{2}} \{+c\}$$

One power correct

M1

Condone absence of "+ c"

A1,A1

3

(ii) Area under curve =

$$4 \frac{4^{\frac{3}{2}}}{1.5} - \frac{4^{\frac{5}{2}}}{2.5} - \{0\}$$

$F(4) - \{F(0)\}$

M1

Total area = $F(4)$ + area triangle OPQ

m1

$$\text{Total area} = \frac{128}{15} + 4 = \frac{188}{15} = 12.5(3\dots)$$

Accept 3 sf if clear

A1

3

[20]

Q35.

(a) (i)

$$t = \frac{1}{2} \quad x = 2 \times \frac{1}{2} + \frac{1}{\left(\frac{1}{2}\right)^2} \quad y = 2 \times \frac{1}{2} - \frac{1}{\left(\frac{1}{2}\right)^2}$$

M1

$$x = 5 \qquad y = -3$$

A1

2

$$(ii) \quad \frac{dy}{dt} = 2 + 2t^{-3} \quad \frac{dx}{dt} = 2 - 2t^{-3}$$

$$2 \text{ and } \frac{d}{dt} \left(\frac{1}{t^2} \right) \text{ attempted in both derivatives}$$

M1A1

$$t = \frac{1}{2} \quad \frac{dy}{dx} = \frac{2 + \frac{2}{\cancel{1/8}}}{2 - \frac{2}{\cancel{1/8}}} = -\frac{9}{7}$$

use chain rule; expressions can be in terms of t or evaluated

M1

CAO or any equivalent fraction (not decimals)

A1

$$y + 3 = -\frac{9}{7}(x - 5)$$

ft on, x y and gradient

if $y = mx + c$ used, c must be found correctly and the equation must be rewritten

B1F

5

$$(b) \quad x - y = \frac{2}{t^2} \quad x + y = 4t$$

either correct expression

$$\text{or both of } x - y = 4t \text{ and } x + y = \frac{2}{t^2}$$

M1

$$\frac{2}{(x - y)} = \left(\frac{x + y}{4} \right)^2$$

eliminate t

M1

$$32 = (x - y)(x + y)^2$$

$$\text{or } (x - y)(x + y)^2 = \frac{2}{t^2} \times (4t)^2 = 32$$

$k = 32$ alone, no marks

A1

3

[10]



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